TRANSFORM CALCULUS, FOURIER SERIES AND NUMERICAL TECHNIQUES

(Course Code: 23MA3BSTFN)

UNIT 1 - LAPLACE TRANSFORMS

Laplace Transform of $f(t)$ denoted by $L\left(f(t)\right)$ is defined by $L\left(f(t)\right)=\int_0^\infty e^{-st} f(t)\,dt = F(s)$ provided that the integral exists where s is a parameter which may be a real or complex number.

Transforms of Elementary functions

1. $L(k)=\dfrac{k}{s}$, $s>0$, where k is a constant
2. $L(t^n) = \begin{cases} \frac{n!}{s^{n+1}} \text{ when } n \text{ is } a \text{ positive integer} \\ \frac{\Gamma(n+1)}{s^{n+1}} \text{ for all n, except for negative integers and } s>0 \end{cases}$
3. $L\left(e^{+at}\right)=\dfrac{1}{s-a}$, $s>a$ and $L\left(e^{-at}\right)=\dfrac{1}{s+a}$, $s>-a$
4. $L(\sin at)=\dfrac{a}{s^2+a^2}$
5. $L(\cos at)=\dfrac{s}{s^2+a^2}$
6. $L(\cosh at)=\dfrac{s}{s^2-a^2}$
7. $L(\sinh at)=\dfrac{a}{s^2-a^2}$

Properties of Laplace transforms (Statement only)

1. **Linearity property**: If $L\left(f(t)\right)=F(s)$, $L\left(g(t)\right)=G(s)$ and $L\left(h(t)\right)=H(s)$ and let *a, b, c* be any constants then $L\left(af(t)+bg(t)-ch(t)\right)=aF(s)+bG(s)-cH(s)$.
2. **Shifting property**: If $L\left(f(t)\right)=F(s)$, then $L\left(e^{at} f(t)\right)=F(s-a)$
3. **Multiplication by t**: If $L\left(f(t)\right)=F(s)$, then $L\left(t\,f(t)\right)=-\dfrac{d}{ds}\left(F(s)\right)$.

   In general, $L\left(t^n\,f(t)\right)=(-1)^n\dfrac{d^n}{ds^n}\left(F(s)\right)$ *for* $n=1,2,3,.....$
4. **Division by t**: If $L\left(f(t)\right)=F(s)$, then $L\left(\dfrac{f(t)}{t}\right)=\int_s^\infty F(s)\,ds$.
5. **Laplace transform of derivatives:** If $L\left(f(t)\right)=F(s)$, then $L\left(f'(t)\right)=sF(s)-f(0)$.

   In general, $L\left(f^n(t)\right)=s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)-.....-f^{n-1}(0)$
6. **Laplace transform of integral:** If $L\left(f(t)\right)=F(s)$, then $L\left(\int_0^t f(u)\,du\right)=\dfrac{F(s)}{s}$, $s>0$.

Find the Laplace transform for the following function $f(t)$ using the definition.

1. $f(t)=\begin{cases}\cos(t-2\pi/3) & t>2\pi/3\\ 0 & t<2\pi/3\end{cases}$ $\quad Ans:\ e^{-2\pi s/3}\left[\dfrac{s}{s^2+1}\right]$

2. $f(t)=\begin{cases}t^2 & 0<t<2\\ t-1 & 2<t<3\\ 7 & t>3\end{cases}$ $\quad Ans:\ \dfrac{2}{s^3}-\dfrac{e^{-2s}}{s^3}\left(2+3s+3s^2\right)+\dfrac{e^{-3s}}{s^2}\left(5s-1\right)$

3. $f(t)=\begin{cases}\sin 2t & 0<t<\pi\\ 0 & t>\pi\end{cases}$ $\quad Ans: \frac{2(1+e^{-\pi s})}{s^2+4}$

4. $f(t)=\begin{cases}e^t & 0<t<1\\ 0 & t>1\end{cases}$. $\quad Ans: \frac{1-e^{-(s-1)}}{s-1}$

Find the Laplace transform for the following function $f(t)$ using linearity property.

1. $f(t)=4e^{5t}+6t^3-3\sin 4t+2\cos 2t$ $\quad$ Ans: $\dfrac{4}{s-5}+\dfrac{36}{s^4}-\dfrac{12}{s^2+16}+\dfrac{2s}{s^2+4}$

2. $f(t)=\cos 3t\cos 2t\cos t$ $\quad$ Ans: $\dfrac{1}{4}\left(\dfrac{s}{s^2+32}+\dfrac{s}{s^2+16}+\dfrac{s}{s^2+4}+\dfrac{1}{s}\right)$

3. $f(t)=3\sinh 5t-4\cosh 5t$ $\quad$ Ans: $\dfrac{3-4s}{s^2-25}$.

4. $f(t)=\cos^3 at$ $\quad$ Ans: $s\left(\frac{1}{s^2+9a^2}+\frac{3}{s^2+a^2}\right)$

5. $f(t)=t\sqrt{t}+4t^3+3^t$ $\quad$ Ans: $3\sqrt{\frac{\pi}{16s^5}}+\frac{24}{s^4}+\frac{1}{s-\log 3}$

6. $f(t)=\sin^3 2t$ $\quad$ Ans: $\frac{3}{4(s^2+1)}-\frac{9}{4(s^2+9)}$

Find the Laplace transform for the following function $f(t)$ using shifting property.

1. $f(t)=(t+2)^2e^t$ $\quad$ Ans: $\dfrac{4s^2-4s+2}{(s-1)^3}$

2. $f(t)=e^{4t}\sin 2t\cos t$ $\quad$ Ans: $\dfrac{1}{2}\left(\dfrac{3}{(s-4)^2+9}+\dfrac{1}{(s-4)^2+1}\right)$

3. $f(t)=e^{-4t}t^{5/2}$ $\quad$ Ans: $\frac{\Gamma\left(\frac{7}{2}\right)}{(s+4)^{\frac{7}{2}}}$

4. $f(t)=(1+te^{-t})^3$ $\quad$ Ans: $\dfrac{1}{s}+\dfrac{3}{(s+1)^2}+\dfrac{6}{(s+2)^3}+\dfrac{6}{(s+3)^4}$

5. $f(t)=e^{-4t}\cosh 2t$ $\quad$ Ans: $\dfrac{s+4}{s^2+8s+12}$.

6. $f(t)=\sinh at\sin at$ $\quad$ Ans: $\dfrac{2a^2s}{\left(s^4+4a^4\right)}$.

Find the Laplace transform for the following function using Multiplication by t property.

1. $f(t) = t\sin^2 t$ Ans: $\dfrac{2(3s^2+4)}{s^2\left(s^2+4\right)^2}$

2. $f(t) = \sin 2t - 2t\cos 2t$ Ans: $\dfrac{16}{\left(s^2+4\right)^2}$

3. $f(t) = te^{2t}\sin 3t$ Ans: $\dfrac{6(s-2)}{(s^2-4s+13)^2}$

4. $f(t) = t^2\sin t$ Ans: $\frac{2(s^2-1)}{(s^2+1)^2}$

5. $f(t) = te^{-3t}cos2t$ Ans: $\frac{(s^2+6s+5)}{(s^2+6s+13)^2}$

6. $f(t) = tsin3tcost$ Ans: $\frac{4s}{(s^2+16)^2} + \frac{2s}{(s^2+4)^2}$

Find the Laplace transform for the following function using division by t property.

1. $f(t) = \dfrac{e^{-at} - e^{-bt}}{t}$ Ans: $\log\left(\dfrac{s+b}{s+a}\right)$

2. $f(t) = \dfrac{\sin^2 t}{t}$ Ans: $\dfrac{1}{4}\log\left(\dfrac{s^2+4}{s^2}\right)$

3. $f(t) = 2^t + \dfrac{\cos at - \cos bt}{t} + t\sin t$ Ans: $\dfrac{1}{s-\log 2} + \dfrac{1}{2}\log\left(\dfrac{s^2+b^2}{s^2+a^2}\right) + \dfrac{s^2-1}{\left(s^2+1\right)^2}$

4. $f(t) = \dfrac{\sin 3t\cos t}{t}$ Ans: $\dfrac{1}{2}\left[\pi - \tan^{-1}\left(\dfrac{s}{4}\right) - \tan^{-1}\left(\dfrac{s}{2}\right)\right]$

5. $f(t) = \dfrac{e^{-t}\sin t}{t}$ Ans: $\cot^{-1}\left(s+1\right)$

Evaluate the following using Laplace transform.

1. $L\left(e^{-t}\int_0^t \dfrac{\sin t}{t}dt\right)$ Ans: $\dfrac{\cot^{-1}\left(s+1\right)}{s+1}$

2. $L\left(t\int_0^t \dfrac{e^{-t}\sin t}{t}dt\right)$ Ans: $\dfrac{s+(s^2+2s+2)\cot^{-1}\left(s+1\right)}{s^2(s^2+2s+2)}$

3. $L\left(\int_0^t \dfrac{1-e^{-t}}{t}dt\right)$ Ans: $\dfrac{1}{s}\log\left(\dfrac{s+1}{s}\right)$

4. $L\left(\int_0^t\int_0^t\int_0^t t\sin t\,dt\,dt\,dt\right)$ Ans: $\dfrac{2}{s^2\left(s^2+1\right)^2}$

5. $L\left(\int_0^t sint\ sin2t\ sin3t\ dt\right)$ Ans: $\frac{1}{4s}\left(\frac{4}{s^2+16} + \frac{2}{s^2+4} - \frac{6}{s^2+36}\right)$

Periodic function: A function $f(t)$ is said to be periodic of period T if $f(t+T) = f(t)$ for $T > 0$.

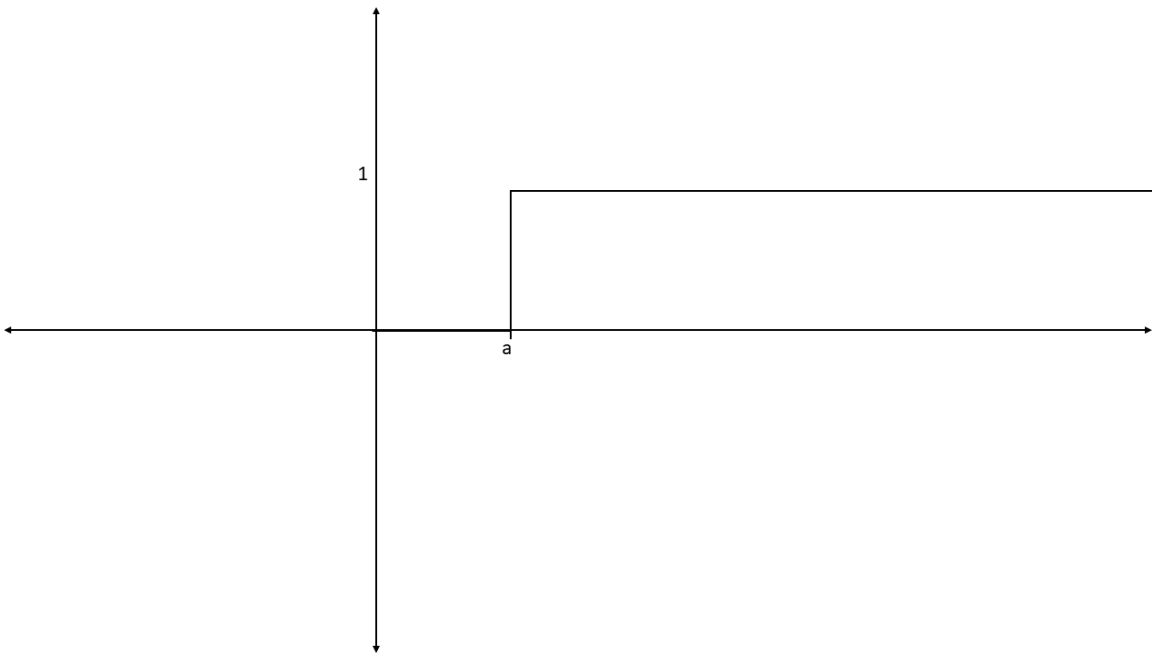
Laplace transforms of a periodic function. (statement only)

If $f(t)$ is a periodic function with period $T>0$ then $L[f(t)]=\dfrac{1}{1-e^{-sT}}\int_0^T e^{-st}f(t)dt$

1. Find the Laplace transform of the square wave function of period a defined as $f(t)=\begin{cases}1 & when\ 0<t<a/2\\ -1 & when\ a/2<t<a\end{cases}$. $Ans:\ \dfrac{1}{s}\tanh\left(\dfrac{as}{4}\right)$
2. Find the Laplace transform of the saw-toothed wave of period T, given $f(t)=\dfrac{t}{T}\ \ for\ \ 0<t<T$. $Ans:\ \dfrac{1}{s^2T}-e^{-sT}\dfrac{(1-e^{sT})}{s}$
3. Find the Laplace transform of the triangular wave function of period $2a$ given by $f(t)=\begin{cases}t & 0<t<a\\ 2a-t & a<t<2a\end{cases}$. $Ans:\ \dfrac{1}{s^2}\tanh\left(\dfrac{as}{2}\right)$
4. Find the Laplace transform of the full wave rectifier $f(t)=E\sin\omega t$, $0<t<\dfrac{\pi}{\omega}$ having period $\dfrac{\pi}{\omega}$. $Ans:\ \dfrac{E\omega}{s^2+\omega^2}\coth\left(\dfrac{\pi s}{2\omega}\right)$
5. Find the Laplace transform of the half wave rectifier $f(t)=\begin{cases}\sin\omega t & 0<t<\pi/\omega\\ 0 & \pi/\omega<t<2\pi/\omega\end{cases}$. $Ans:\ \dfrac{\omega}{\left(1-e^{-\pi s/\omega}\right)\left(s^2+\omega^2\right)}$
6. Find the Laplace transform of the periodic function $f(t)=\begin{cases}1 & 0<t<1\\ 0 & 1<t<2\\ -1 & 2<t<3\end{cases}$ having period 3. $Ans:\dfrac{1-e^{-2s}}{s(1+e^{-s}+e^{-2s})}$
7. Find the Laplace transform of the periodic function $f(t)=\begin{cases}t & 0<t<\pi\\ 0 & \pi<t<2\pi\end{cases}$ having period 2π. $Ans:\dfrac{1-e^{-\pi s}(1+2\pi s)}{s^2(1-e^{-2\pi s})}$
8. Find the Laplace transform of the periodic function $f(t)=\begin{cases}1+t & 0<t\le 1\\ 3-t & 1\le t<2\end{cases}$ having period 2. $Ans:\dfrac{1}{s}-\dfrac{e^{-s}}{s^2(1+e^{-s})}$
9. Find the Laplace transform of the periodic function $f(t)=t^2$, $0<t<\alpha$ and $f(t+\alpha)=f(t)$. $Ans:\dfrac{2}{s^3}-\dfrac{\alpha e^{-\alpha s}(2+\alpha s)}{(1-e^{-\alpha s})s^2}$

Unit step function or Heaviside function: The unit step function $u(t-a)$ is defined as follows:

$u(t-a)=\begin{cases}0 & for\ \ t<a\\ 1 & for\ \ t\ge a\end{cases}$ where a is always positive.



Laplace Transform of unit step function: $L\left[u(t-a)\right]=\dfrac{e^{-as}}{s}$

Second shifting property: If $L\left(f(t)\right)=F(s)$, then $L\left(f(t-a)\cdot u(t-a)\right)=e^{-as}F(s)$ (statement only)

Representation of a function in terms of Heaviside function

If $f(t)=\begin{cases} f_1(t), 0<t<a \\ f_2(t), \quad t>a \end{cases}$, then we can write $f(t)=f_1(t)+\left(f_2(t)-f_1(t)\right)u(t-a)$

Also, it can be extended further as,

If $f(t)=\begin{cases} f_1(t), 0<t<a \\ f_2(t), a<t<b \\ f_3(t), \quad t>b \end{cases}$, then we can write

$$f(t)=f_1(t)+\left(f_2(t)-f_1(t)\right)u(t-a)+\left(f_3(t)-f_2(t)\right)u(t-b)$$

Express the following functions with an interval of the type $(0,\alpha)$ in terms of unit step function. Hence find their Laplace transform.

1. $f(t)=\begin{cases} 2t & 0<t<\pi \\ 1 & t>\pi \end{cases}$ $\quad Ans:\ \dfrac{2}{s^2}+e^{-as}\left(\dfrac{1-2\pi}{s}-\dfrac{2}{s^2}\right)$

2. $f(t)=\begin{cases} \cos t & 0<t<\pi \\ \sin t & t>\pi \end{cases}$ $\quad Ans: \dfrac{s(1+e^{-\pi s})-e^{-\pi s}}{s^2+1}$

3. $f(t)=\begin{cases} e^{-t} & 0<t<3 \\ 0 & t>3 \end{cases}$ $\quad Ans: \dfrac{1-e^{-3s}}{s+1}$

4. $f(t)=\begin{cases} \sin t & 0<t<\pi \\ \sin 2t & \pi<t<2\pi \\ \sin 3t & t>2\pi \end{cases}$ $\quad Ans:\ \dfrac{1}{s^2+1}+e^{-\pi s}\left(\dfrac{2}{s^2+4}-\dfrac{1}{s^2+1}\right)+e^{-2\pi s}\left(\dfrac{3}{s^2+9}-\dfrac{2}{s^2+4}\right)$

5. $f(t)=\begin{cases}\cos t & for\ \ 0<t\le\pi\\ 1 & for\ \ \pi<t\le 2\pi\\ \sin t & for\ \ t>2\pi\end{cases}$ $\quad Ans:\ \dfrac{s}{s^2+1}+e^{-\pi s}\left(\dfrac{s}{s^2+1}+\dfrac{1}{s}\right)-e^{-2\pi s}\left(\dfrac{1}{s}-\dfrac{1}{s^2+1}\right)$

6. $f(t)=\begin{cases}t^2 & 0<t<2\\ 4t & 2<t<4\\ 8 & t>4\end{cases}$ $\quad Ans:\ \dfrac{2}{s^3}\left\{1+e^{-2s}\left(2s^2-1\right)-2e^{-4s}\left(1+4s\right)\right\}$

7. $f(t)=\begin{cases}t^2 & 0<t\le 2\\ 0 & t>2\end{cases}$ $\quad Ans:\ \dfrac{2(1-e^{-2s})}{s^3}-\dfrac{4e^{-2s}(1+s)}{s^2}$

8. $f(t)=\begin{cases}0 & 0<t<5\\ t-3 & t>5\end{cases}$ $\quad Ans:\ -\frac{e^{-5s}}{s^2}(1+2s)$

9. $f(t)=\begin{cases}2 & 0<t<\pi\\ 0 & \pi<t<2\pi\\ \sin t & t>2\pi\end{cases}$. $\quad Ans: \frac{e^{-2\pi s}}{s^2+1}-\frac{2}{s}(1+e^{-\pi s})$

<u>Inverse Laplace Transforms</u>

If $L\left(f(t)\right)=F(s)$ then Inverse Laplace Transform of $F(s)$ is denoted by $L^{-1}\left(F(s)\right)$ and is given as $L^{-1}\left(F(s)\right)=f(t)$.

Inverse Laplace Transforms of Elementary functions

1. $L^{-1}\left(\dfrac{1}{s}\right)=1$, $s>0$
2. $L^{-1}\left(\frac{1}{s^{n+1}}\right)=\frac{t^n}{n!}$, when n is a positive integer
3. $L^{-1}\left(\dfrac{1}{s-a}\right)=e^{+at}$
4. $L^{-1}\left(\dfrac{1}{s^2+a^2}\right)=\dfrac{\sin at}{a}$
5. $L^{-1}\left(\dfrac{s}{s^2+a^2}\right)=\cos at$
6. $L^{-1}\left(\dfrac{s}{s^2-a^2}\right)=\cosh at$
7. $L^{-1}\left(\dfrac{1}{s^2-a^2}\right)=\dfrac{\sinh at}{a}$

Properties of Inverse Laplace transforms

1. **Linearity property**: If $L^{-1}\left(F(s)\right)=f(t)$, $L^{-1}\left(G(s)\right)=g(t)$ and $L^{-1}\left(H(s)\right)=h(t)$ and let a, b, c be any constants then $L^{-1}\left(aF(s)+bG(s)-cH(s)\right)=af(t)+bg(t)-ch(t)$.
2. **Shifting property**: If $L^{-1}\left(F(s)\right)=f(t)$, then $L^{-1}\left(F(s-a)\right)=e^{+at}f(t)$.
3. **Second Shifting property:** If $L^{-1}\left(F(s)\right)=f(t)$ then $L^{-1}\left(e^{-as}F(s)\right)=f(t-a)\cdot u(t-a)$.

Evaluation of $L^{-1}\{F(S)\}$:

1. $\left[\frac{2}{s}-\frac{1}{s^3}\right]^2$ $\quad Ans: 4t+\frac{t^5}{5!}-4\frac{t^3}{3!}$
2. $\frac{-2s+6}{s^2-4}$ $\quad Ans: -2\cosh 2t+\frac{3\sinh 2t}{2}$
3. $\frac{(s+1)^3}{s^4}$ $\quad Ans: 1+\frac{t^3}{6}+3t+\frac{3t^2}{2}$
4. $\frac{4s}{4s^2+1}+\frac{1}{5s^2+1}$ $\quad Ans: cos(1/2)t+\frac{1}{\sqrt{5}}sin(1/\sqrt{5})t$
5. $\frac{1}{4s+1}+\frac{1}{5s-2}$ $\quad Ans: \frac{e^{-(1/4)t}}{4}+\frac{e^{(2/5)t}}{5}$

Evaluation of $L^{-1}\{F(s-a)\}$

1. $\frac{1}{(s+2)^3}$ $\quad Ans: \frac{e^{-2t}t^2}{2}$
2. $\frac{2s-3}{s^2+4s+13}$ $\quad Ans: 2e^{-2t}\cos 3t-\frac{7e^{-2t}\sin 3t}{3}$
3. $\frac{s}{s^2+4s+5}$ $\quad Ans: e^{-2t}\cos t-2e^{-2t}\sin t$
4. $\frac{2s+5}{s^2+6s+34}$ $\quad Ans: 2e^{-3t}\cos 5t-\frac{e^{-3t}\sin 5t}{5}$
5. $\frac{s^2}{(s-a)^3}$ $\quad Ans: \frac{a^2e^{at}t^2}{2}+2ate^{at}+e^{at}$
6. $\frac{4}{(s-4)^5}$ $\quad Ans: \frac{4e^{4t}t^4}{4!}$
7. $\frac{2s+5}{(s-3)^2}$ $\quad Ans: 2e^{3t}+11e^{3t}t$
8. $\frac{1}{s^2+2s+5}$ $\quad Ans: \frac{e^{-t}\sin 2t}{2}$
9. $\frac{1}{(s+4)^{5/2}}$ $\quad Ans: \frac{4e^{-4t}t^{(3/2)}}{3\sqrt{\pi}}$

Evaluation of $L^{-1}\{F(S)\}$ using partial fraction:

1. $\frac{s^2+6s+9}{(s-1)(s-2)(s+4)}$ $\quad Ans: \frac{16e^t}{5}-e^{2t}+\frac{49e^{-4t}}{30}$
2. $\frac{s+1}{s^2-4s}$ $\quad Ans: \frac{-1}{4}+\frac{5e^{4t}}{4}$
3. $\frac{s^2+1}{s(s-1)(s-2)(s+1)}$ $\quad Ans: \frac{1}{2}-e^t+\frac{5e^{2t}}{6}-\frac{1}{3}e^{-t}$

4. $\dfrac{(s+1)^2}{(s+2)^4}$ $\quad Ans: \dfrac{-e^{-2t}t^3}{6} + e^{-2t}t^2 + e^{-2t}t$

5. $\dfrac{s}{(s+2)(s^2+4)}$ $\quad Ans: \dfrac{cos2t+sin2t-e^{-2t}}{4}$

6. $\dfrac{s}{(s-2)(s-3)(s-6)}$ $\quad Ans: \dfrac{e^{2t}}{2} - e^{3t} + e^{6t}$

7. $\dfrac{2s-1}{s^2(s+1)^3}$ $\quad Ans: 1 - t + e^{-t}\left(4 - 5t + \dfrac{t^2}{2}\right)$

Evaluation of $L^{-1}\left\{e^{-as}F(s)\right\}$:

1. $\dfrac{e^{-3s}}{(s-4)^2}$ $\quad Ans: \left(e^{4(t-3)}(t-3)\right)u(t-3)$

2. $\dfrac{s\,e^{-\pi s}}{s^2+9}$ $\quad Ans: -\cos 3t\ u(t-\pi)$

3. $\dfrac{s\,e^{-s/2}+\pi e^{-s}}{s^2+\pi^2}$ $\quad Ans: \sin \pi t\ u\left[t-\dfrac{1}{2}\right] - \sin \pi t\, u(t-1)$

4. $\dfrac{s\,e^{-as}}{s^2-k^2}$ $\quad Ans: \cosh k(t-a)u(t-a)$

5. $\dfrac{e^{-3s}}{(s+4)^{5/2}}$ $\quad Ans: \dfrac{4e^{-4(t-3)}(t-3)^{(3/2)}}{3\sqrt{\pi}}u(t-3)$

6. $\dfrac{\cosh 2s}{e^{3s}s^2}$ $\quad Ans: \dfrac{(t-1)u(t-1)+(t-3)u(t-3)}{2}$

7. $\dfrac{e^{-2s}}{s(s^2-1)}$ $\quad Ans: u(t-2)(\cosh(t-2)-1)$

8. $\dfrac{e^{-\pi s}}{s^2+1}$ $\quad Ans: -\sin t\ u(t-\pi)$

Application to Differential equations

Solve the following DE by the method of Laplace transform.

1. $y''' + 2y'' - y' - 2y = 0$ given $y(0) = y'(0) = 0$ and $y''(0) = 6$. $Ans: e^t - 3e^{-t} + 2e^{-2t}$

2. $(D^3 - 3D^2 + 3D - 1)y = t^2e^t$ given $y(0) = 1$, $y'(0) = 0$ and $y'''(0) = 1$.

   $Ans: e^t\left\{\dfrac{t^5}{60} + \dfrac{t^2}{2} + 1 - t\right\}$

3. $\dfrac{d^2x}{dt^2} + 9x = \cos(2t)$ if $x(0) = 1$ and $x\left(\pi/2\right) = -1$. $\quad Ans: x = \dfrac{cos2t+4cos3t+4sin3t}{5}$

4. $\dfrac{d^2y}{dx^2} + y = 0$ if $y(0) = 2$ and $y\left(\pi/2\right) = 1$. $\quad Ans: y = 2cosx + sinx$

5. $(D^2 - 1)x = a\cosh t$ given $x(0) = 0 = x'(0)$. $\quad Ans: x = \dfrac{atsinht}{2}$

6. $(D^2 - D)y = e^t\cos t$ $y(0) = 0$ $y'(0) = 0$ $\quad Ans: \frac{1}{2}\{1 + e^t(\sin t - \cos t)\}$

7. $\dfrac{d^2x}{dt^2} - 2\dfrac{dx}{dt} + x = e^t$ with $x = 2$, $\dfrac{dx}{dt} = -1$, at $t = 0$.

   $Ans: e^t\left\{2 - 3t + \frac{t^2}{2}\right\}$